CHAPTER 9

COUNTING AND PROBABILITY

9.9 Conditional Probability, Bayes' Formula, and Independent Events

Conditional Probability

Imagine a couple with two children, each of whom is equally likely to be a boy or a girl. Now suppose you are given the information that one is a boy. What is the probability that the other child is a boy?

Figure 9.9.1 shows the four equally likely combinations of gender for the children.

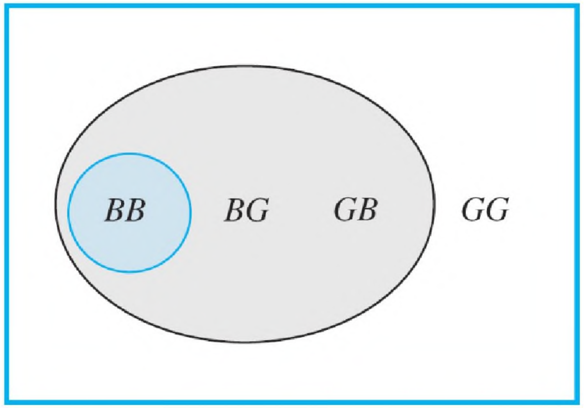


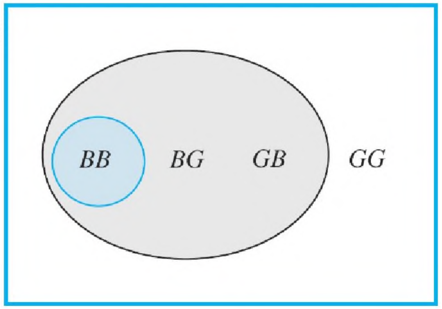
Figure 9.9.1

Conditional Probability

Within the new sample space, there is one combination where the other child is a boy (in the region shaded blue-gray).

Thus, it would be reasonable to say that the likelihood that the other child is a boy, given that at least one is a boy, is $1/3 = 33\frac{1}{3}$ %.   Also,

Note that because the original sample space co outcomes,



$$\frac{P(\text{at least one child is a boy and the other child is also a boy})}{P(\text{at least one child is a boy})} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

Conditional Probability

Definition

Let A and B be events in a sample space S. If $P(A) \neq 0$, then the **conditional probability of B given A**, denoted $\boldsymbol{P(B|A)}$, is

$$P(B|A) = \frac{P(A \cap B)}{P(A)}. \qquad 9.9.1$$

Example 9.9.1 – *Computing a Conditional Probability*

A pair of fair dice, one blue and the other gray, are rolled. What is the probability that the sum of the numbers showing face up is 8, given that both of the numbers are even?

Example 9.9.1 – *Solution*

The sample space is the set of all 36 outcomes obtained from rolling the two dice and noting the numbers showing face up on each.

As in earlier section, denote by *ab* the outcome that the number showing face up on the blue die is *a* and the one on the gray die is *b*.

Let *A* be the event that both numbers are even and *B* the event that the sum of the numbers is 8.

Example 9.9.1 – *Solution*

continued

Then $A = \{22, 24, 26, 42, 44, 46, 62, 64, 66\}$, $B = \{26, 35, 44, 53, 62\}$, and $A \cap B = \{26, 44, 62\}$.

Because the dice are fair (so all outcomes are equally likely), $P(A) = 9/36$, $P(B) = 5/36$, and $P(A \cap B) = 3/36$. By definition of conditional probability,

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{3}{36}}{\frac{9}{36}} = \frac{3}{9} = \frac{1}{3}.$$

Conditional Probability

Note that when both sides of the formula for conditional probability (formula 9.9.1) are multiplied by $P(A)$, a formula for $P(A \cap B)$ is obtained:

$$P(A \cap B) = P(B|A) \cdot P(A). \qquad 9.9.2$$

And dividing both sides of formula (9.9.2) by $P(B|A)$ gives a formula for $P(A)$:

$$P(A) = \frac{P(A \cap B)}{P(B|A)}. \qquad 9.9.3$$

Example 2 – *Representing Conditional Probabilities with a Tree Diagram*

An urn contains 5 blue and 7 gray balls. Let us say that 2 are chosen at random, one after the other, without replacement.

a. Find the following probabilities and illustrate them with a tree diagram: the probability that both balls are blue, the probability that the first ball is blue and the second is not blue, the probability that the first ball is not blue, and the second ball is blue, and the probability that neither ball is blue.

b. What is the probability that the second ball is blue?

c. What is the probability that at least one of the balls is blue?

d. If the experiment of choosing two balls from the urn were repeated many times over, what would be the expected value of the number of blue balls?

Solution:

Let S denote the sample space of all possible choices of two balls from the urn, let $B1$ be the event that the first ball is blue, and let $B2$ be the event that the second ball is blue.

Example 2 – *Solution*

cont'd

Then B_1^c is the event that the first ball is not blue and B_2^c is the event that the second ball is not blue.

a. Because there are 12 balls of which 5 are blue and 7 are gray, the probability tl ball is blue is

$$P(B_1) = \frac{5}{12}$$

and the probability that the first ball is not blue is

$$P(B_1^c) = \frac{7}{12}.$$

Example 2 – *Solution*

cont'd

If the first ball is blue, then the urn would contain 4 blue balls and 7 gray balls, and so

$$P(B_2 \mid B_1) = \frac{4}{11} \quad \text{and} \quad P(B_2^c \mid B_1) = \frac{7}{11},$$

where $P(B_2 \mid B_1)$ is the probability that tl $P(B_2^c \mid B_1)$ ball is blue given that the first ball is blue and is the probability that the second ball is not blue given that the first ball is blue.

Example 2 – *Solution*

cont'd

It follows from formula (9.9.2)

$$\boxed{P(A \cap B) = P(B \mid A) \cdot P(A).} \qquad 9.9.2$$

that

$$P(B_1 \cap B_2) = P(B_2 \mid B_1) \cdot P(B_1) = \frac{4}{11} \cdot \frac{5}{12} = \frac{20}{132}$$

and

$$P(B_1 \cap B_2^c) = P(B_2^c \mid B_1) \cdot P(B_1) = \frac{7}{11} \cdot \frac{5}{12} = \frac{35}{132}.$$

Example 2 – *Solution*

cont'd

Similarly, if the first ball is not blue, then the urn would contain 5 blue balls and 6 gray balls, and so

$$P(B_2 \mid B_1^c) = \frac{5}{11} \quad \text{and} \quad P(B_2^c \mid B_1^c) = \frac{6}{11},$$

where $P(B_2 \mid B_1^c)$ is the probability that the second ball is blue given that the first ball is not blue and $P(B_2^c \mid B_1^c)$ is the probability that the second ball is not blue given that the first ball is not blue.

Example 2 – *Solution*

cont'd

It follows from formula (9.9.2) that

$$P(B_1^c \cap B_2) = P(B_2 \mid B_1^c) \cdot P(B_1^c) = \frac{5}{11} \cdot \frac{7}{12} = \frac{35}{132}$$

and

$$P(B_1^c \cap B_2^c) = P(B_2^c \mid B_1^c) \cdot P(B_1^c) = \frac{6}{11} \cdot \frac{7}{12} = \frac{42}{132}.$$

Example 2 – *Solution*

cont’d

The tree diagram in Figure 9.9.2 is a convenient way to

help calculate these results

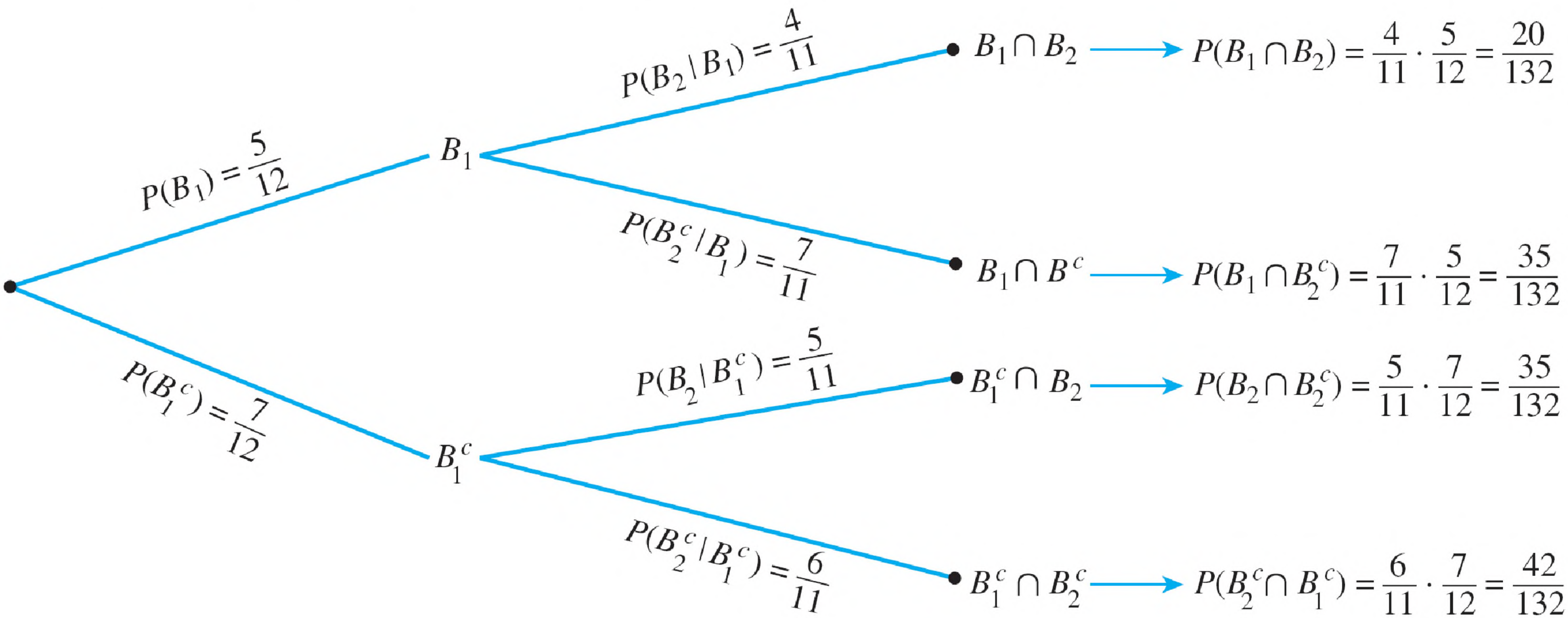


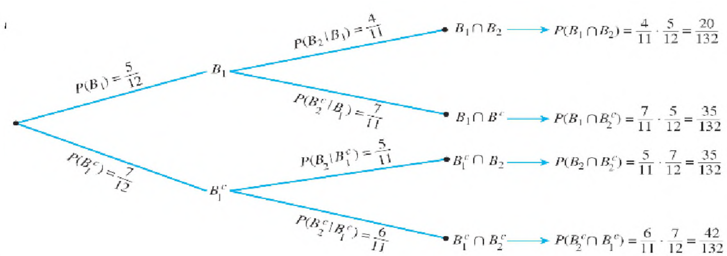
Figure 9.9.2

Example 2 – *Solution*

cont'd

b. What is the probability that the second ball is blue?

b. The event that the second ball is blue can occur in one of two mutually exclusive ways: Either the first ball is blue and the second is also blue, or the first ball is gray and the second is blue. In other words, *B*2 is the disjoint union of *B*2 ∩ *B*1 and *B*2 ∩ B_1^c.



Hence

$$
\begin{aligned}
P(B_2) &= P\left((B_2 \cap B_1) \cup \left(B_2 \cap B_1^c\right)\right) \\
&= P(B_2 \cap B_1) + P\left(B_2 \cap B_1^c\right) && \text{by probability axiom 3} \\
&= \frac{20}{132} + \frac{35}{132} && \text{by part (a)}
\end{aligned}
$$

Example 2 – *Solution*

cont'd

$$= \frac{55}{132} = \frac{5}{12}.$$

Thus, the probability that the second ball is blue is 5/12, the same as the probability that the first ball is blue.

c. What is the probability that at least one of the balls is Blue?

c. By formula 9.8.2, for the union of any two events,

Probability of a General Union of Two Events

If S is any sample space and A and B are any events in S, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \qquad 9.8.2$$

$$P(B_1 \cup B_2) = P(B_1) + P(B_2) - P(B_1 \cap B_2)$$

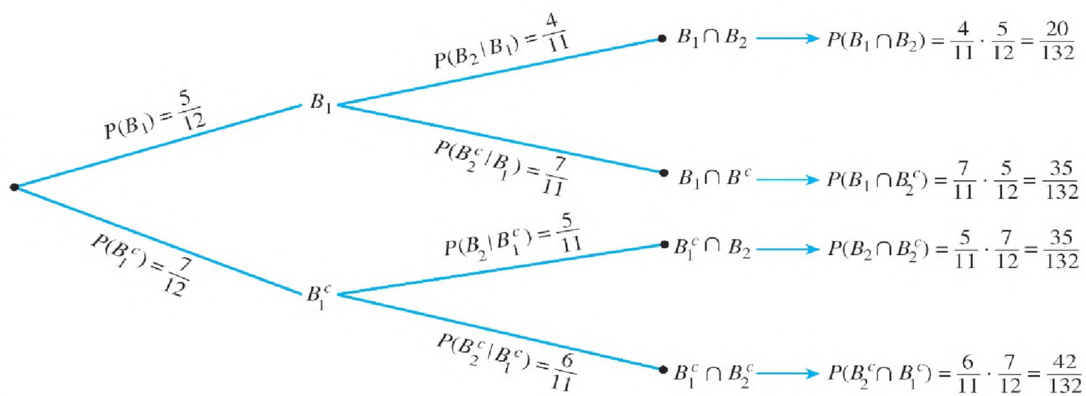
Example 2 – *Solution*

cont'd

$$= \frac{5}{12} + \frac{5}{12} - \frac{20}{132} \qquad \text{by parts (a) and (b)}$$

$$= \frac{90}{132}$$

$$= \frac{15}{22}.$$



Thus, the probability is 15/22, or approximately 68.2%, that at least one of the balls is blue.

Also: the probability that at least one of the balls is Blue = 1- 42/132 = 15/22

Example 2 – *Solution*

cont'd

d. If the experiment of choosing two balls from the urn were repeated many times over, what would be the expected value of the number of blue balls?

d. The event that neither ball is blue is the complement of the event that at least one of the balls is blue, so

$$P(0 \text{ blue balls}) = 1 - P(\text{at least one ball is blue}) \qquad \text{by formula 9.8.1}$$

Probability of the Complement of an Event

If A is any event in a sample space S, then

$$P(A^c) = 1 - P(A). \qquad 9.8.1$$

$$= 1 - \frac{15}{22} \qquad \text{by part (c)}$$

$$= \frac{7}{22}.$$

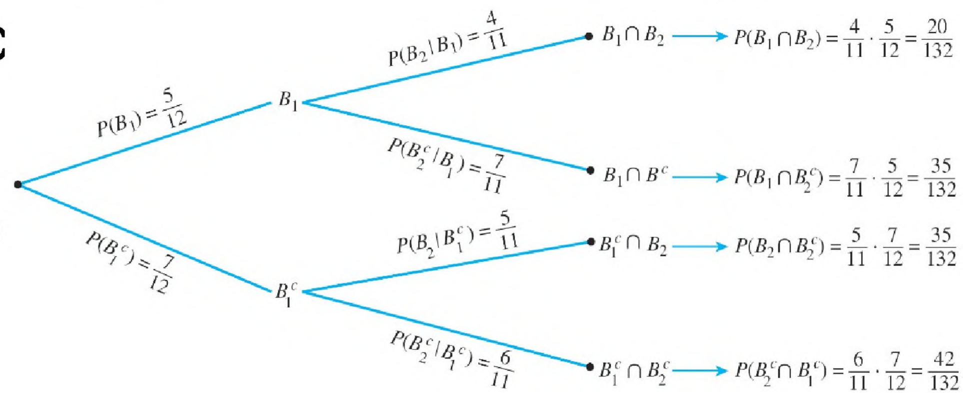
Example 2 – *Solution*

cont'd

The event that one ball is blue can occur in one of two mutually exclusive ways: Either the second ball is blue and the first is not, or the first ball is blue and the second is not.

Part (a) showed that the probability of the first way is $\frac{35}{132}$,

and the probability of the sec



Thus, by probability axiom 3,

$$P(1 \text{ blue ball}) = \frac{35}{132} + \frac{35}{132} = \frac{70}{132}.$$

Example 2 – *Solution*

cont'd

Finally, by part (a),

$$P(2 \text{ blue balls}) = \frac{20}{132}.$$

Therefore,

$$\begin{aligned}\begin{bmatrix}\text{the expected value of} \\ \text{the number of blue balls}\end{bmatrix} &= 0 \cdot P(0 \text{ blue balls}) + 1 \cdot P(1 \text{ blue ball}) \\ &\qquad + 2 \cdot P(2 \text{ blue balls}) \\ &= 0 \cdot \frac{7}{22} + 1 \cdot \frac{70}{132} + 2 \cdot \frac{20}{132} \\ &= \frac{110}{132} \cong 0.8.\end{aligned}$$

Bayes' Theorem

Suppose that one urn contains 3 blue and 4 gray balls, and a second urn contains 5 blue and 3 gray balls. A ball is selected by choosing one of the urns at random and then picking a ball at random from that urn. If the chosen ball is blue, what is the probability that it came from the first urn?

This problem can be solved by carefully interpreting all the information that is known and putting it together in just the right way.

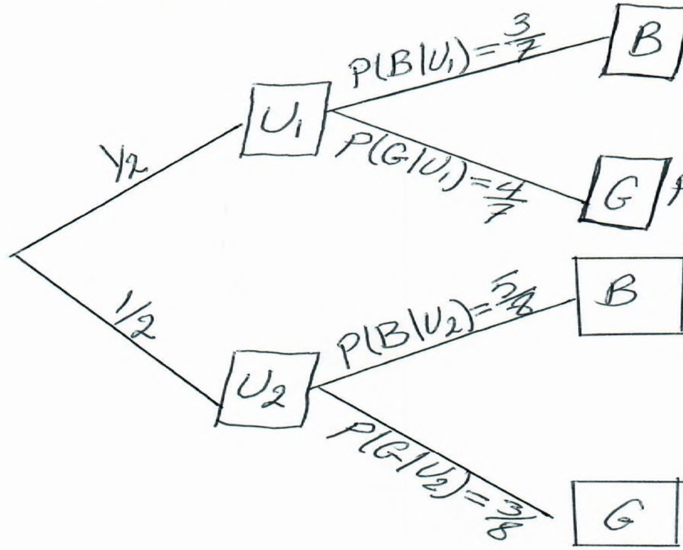
Let A be the event that the chosen ball is blue, $B1$ the event that the ball came from the first urn, and $B2$ the event that the ball came from the second urn.

Bayes' Theorem

$U_1: 3B, 4G$

$U_2: 5B, 3G$

Bayes' Theorem Example



$P(U_1 \cap B) = \frac{3}{7} \cdot \frac{1}{2} = \frac{3}{14}$

$P(U_1 \cap G) = \frac{4}{7} \cdot \frac{1}{2} = \frac{4}{14} = \frac{2}{7}$

$P(U_2 \cap B) = \frac{5}{8} \cdot \frac{1}{2} = \frac{5}{16}$

$P(U_2 \cap G) = \frac{3}{8} \cdot \frac{1}{2} = \frac{3}{16}$

$$P(U_1 | B) = \frac{P(U_1 \cap B)}{P(B)} = \frac{\frac{3}{14}}{\frac{3}{14} + \frac{5}{16}} = \frac{336}{826} \cong 40.7\%$$

$$P(A \cap B) = P(B|A) \cdot P(A)$$

Bayes' Theorem

Because 3 of the 7 balls in urn one are blue, and 5 of the 8 balls in urn two are blue,

$$P(A \mid B_1) = \frac{3}{7} \quad \text{and} \quad P(A \mid B_2) = \frac{5}{8}.$$

And because the urns are equally likely to be chosen,

$$P(B_1) = P(B_2) = \frac{1}{2}.$$

Bayes' Theorem

Moreover, by formula (9.9.2),

$$P(A \cap B) = P(B \mid A) \cdot P(A). \qquad 9.9.2$$

$$P(A \cap B_1) = P(A \mid B_1) \cdot P(B_1) = \frac{3}{7} \cdot \frac{1}{2} = \frac{3}{14}, \quad \text{and}$$

$$P(A \cap B_2) = P(A \mid B_2) \cdot P(B_2) = \frac{5}{8} \cdot \frac{1}{2} = \frac{5}{16}.$$

But A is the disjoint union of ($A \cap B1$) and ($A \cap$ B2), so by probability axiom 3,

$$P(A) = P((A \cap B_1) \cup (A \cap B_2)) = P(A \cap B_1) + P(A \cap B_2) = \frac{3}{14} + \frac{5}{16} = \frac{59}{112}.$$

Bayes' Theorem

Finally, by definition of conditional probability,

$$P(B_1 \mid A) = \frac{P(B_1 \cap A)}{P(A)} = \frac{\frac{3}{14}}{\frac{59}{112}} = \frac{336}{826} \cong 40.7\%.$$

Thus, if the chosen ball is blue, the probability is approximately 40.7% that it came from the first urn.

The steps used to derive the answer in the previous example can be generalized to prove Bayes' Theorem.

Bayes' Theorem

Theorem 9.9.1 Bayes' Theorem

Suppose a sample space S is a union of mutually disjoint events $B_1, B_2, B_3, \ldots, B_n$, suppose A is an event in S, and suppose both A and each B_k have nonzero probabilities for every integer k with $1 \le k \le n$. Then

$$P(B_k|A) = \frac{P(A|B_k)P(B_k)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_n)P(B_n)}.$$

Also:

$$P(B_k|A) = \frac{P(A \cap B_k)}{P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) + \cdots + P(A \cap B_n)}$$

$$P(A \cap B) = P(B|A) \cdot P(A). \quad 9.9.2$$

Example 9.9.3 – *Applying Bayes' Theorem*

Most medical tests occasionally produce incorrect results, called false positives and false negatives.

When a test is designed to determine whether a patient has a certain disease, a **false positive** result indicates that a patient has the disease when the patient does not have it.

A **false negative** result indicates that a patient does not have the disease when the patient does have it.

Example 9.9.3 – *Applying Bayes' Theorem*

continued

Consider a medical test that screens for a disease found in 5 people in 1,000. Suppose that the false positive rate is 3% and the false negative rate is 1%.

Example 9.9.3 – *Applying Bayes' Theorem*

continued

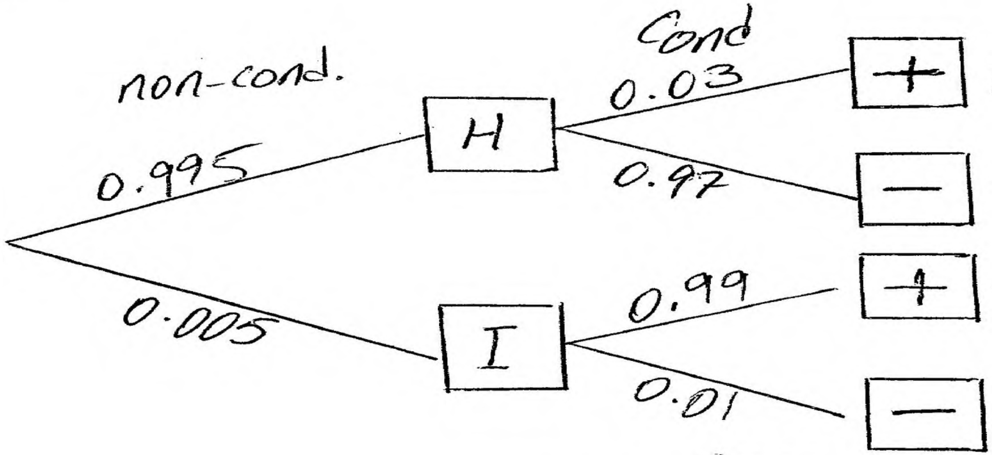
Then 99% of the time a person who has the condition tests positive for it, and 97% of the time a person who does not have the condition tests negative for it.

a. What is the probability that a randomly chosen person who tests positive for the disease actually has the disease?

b. What is the probability that a randomly chosen person who tests negative for the disease does not in fact have the disease?

Example 9.9.3

Given Prob.	Comp. of Given Prob.
$P(H) = 995/1000 = 0.995$	$P(I) = 5/1000 = 0.005$
$P(+\mid H) = 0.03$	$P(-\mid H) = 0.97$
$P(+\mid I) = 0.99$	$P(-\mid I) = 0.01$



$P(H \cap +) = P(+\mid H) \cdot P(H) = (0.03)(0.995)$

$P(H \cap -) = P(-\mid H) \cdot P(H) = (0.97)(0.995)$

$P(I \cap +) = P(+\mid I) \cdot P(I) = (0.99) \cdot (0.005)$

$P(I \cap -) = P(-\mid I) \cdot P(I) = (0.01)(0.005)$

$$P(I \mid +) = \frac{P(I \cap +)}{P(+)} = \frac{P(+\mid I) \cdot P(I)}{P(+\mid I) \cdot P(I) + P(+\mid H) \cdot P(H)}$$

$$= \frac{(0.99)(0.005)}{(0.99)(0.005) + (0.03)(0.995)} = 0.1422 = 14.2\%$$

$$P(H \mid -) = \frac{P(H \cap -)}{P(-)} = \frac{P(-\mid H) \cdot P(H)}{P(-\mid H) \cdot P(H) + P(-\mid I) \cdot P(I)}$$

$$= \frac{(0.97) \cdot (0.995)}{(0.97)(0.995) + (0.01)(0.005)} = 0.99995 = 99.995\%$$

Example 9.9.3 – *Solution*

Consider a person chosen at random from among those screened.

Let *A* be the event that the person tests positive for the disease, *B*1 the event that the person actually has the disease, and *B*2 the event that the person does not have the disease. Then

$$P(A \mid B_1) = 0.99, \quad P(A^c \mid B_1) = 0.01, \quad P(A^c \mid B_2) = 0.97,$$

$$\text{and} \quad P(A \mid B_2) = 0.03.$$

Example 9.9.3 – *Solution*

continued

Also, because 5 people in 1,000 have the disease,

$P(B1) = 0.005$ and $P(B2) = 0.995$.

a. By Bayes' theorem,

$$P(B_1 | A) = \frac{P(A | B_1)P(B_1)}{P(A | B_1)P(B_1) + P(A | B_2)P(B_2)}$$

$$= \frac{(0.99)(0.005)}{(0.99)(0.005) + (0.03)(0.995)}$$

$\cong 0.1422 \cong 14.2\%$.

Example 9.9.3 – *Solution*

continued

Thus the probability that a person with a positive test result actually has the disease is approximately 14.2%.

b. By Bayes' theorem,

$$P(B_2|A^c) = \frac{P(A^c|B_2)P(B_2)}{P(A^c|B_1)P(B_1) + P(A^c|B_2)P(B_2)}$$

$$= \frac{(0.97)(0.995)}{(0.01)(0.005) + (0.97)(0.995)}$$

$$\cong 0.999948 \cong 99.995\%.$$

Example 9.9.3 – *Solution*

continued

Thus the probability that a person with a negative test result does not have the disease is approximately 99.995%.

You might be surprised by these numbers, but they are fairly typical of the situation where the screening test is significantly less expensive than a more accurate test for the same disease yet produces positive results for nearly all people with the disease.

Independent Events

Suppose a fair coin is tossed twice. It seems intuitively clear that the outcome of the first toss does not depend in any way on the outcome of the second toss, and conversely, the outcome of the second toss does not depend on the outcome of the first toss.

In other words, if, for instance, *A* is the event that a head is obtained on the first toss and *B* is the event that a head is obtained on the second toss, then if the coin is tossed randomly both times, events *A* and *B* should be *independent* in the sense that $P(A|B) = P(A)$ and $P(B|A) = P(B)$.

Independent Events

Definition

If A and B are events in a sample space S, then A and B are **independent** if, and only if,

$$P(A \cap B) = P(A) \cdot P(B).$$

Example 9.9.4 – *Disjoint Events and Independence*

Let A and B be events in a sample space S, and suppose $A \cap B = \emptyset$, $P(A) \neq 0$, and $P(B) \neq 0$. Show that

$$P(A \cap B) \neq P(A) \cdot P(B).$$

Example 9.9.4 – *Solution*

Because

$$A \cap B = \emptyset, P(A \cap B) = 0$$

by probability axiom 2. But $P(A) \cdot P(B) \neq 0$ because neither $P(A)$ nor $P(B)$ equals zero.

Thus $P(A \cap B) \neq P(A) \cdot P(B)$.

Example [illegible] – Computing Probabilities of Intersections of Two Independent Events

A coin is loaded so that the probability of heads is 0.6. Suppose the coin is tossed twice.

Although the probability of heads is greater than the probability of tails, there is no reason to believe that whether the coin lands heads or tails on one toss will affect whether it lands heads or tails on the other toss.

Thus, it is reasonable to assume that the results of the tosses are independent.

Example [illegible] – Computing Probabilities of Intersections of Two Independent Events

continued

a. What is the probability of obtaining two heads?

b. What is the probability of obtaining one head?

c. What is the probability of obtaining no heads?

d. What is the probability of obtaining at least one head?

Example 9.9.6 – *Solution*

The sample space S consists of the four outcomes $\{HH, HT, TH, TT\}$, which are not equally likely.

Let E be the event that a head is obtained on the first toss, and let F be the event that a head is obtained on the second toss.

Then $P(E) = P(F) = 0.6$, and it is to be assumed that E and F are independent.

Example 9.9.6 – *Solution*

continued

a. What is the probability of obtaining two heads?

a. The event of obtaining two heads is $E \cap F$. And because E and F are independent,

$$P(\text{two heads}) = P(E \cap F) = P(E) \cdot P(F) = (0.6)(0.6)$$
$$= 0.36 = 36\%.$$

b. What is the probability of obtaining one head?

b. One head can be obtained in two mutually exclusive ways: head on the first toss and tail on the second, or tail on the first toss and head on the second. Thus, the event of obtaining exactly one head is $(E \cap F^c) \cup (E^c \cap F)$.

Example 9.9.6 – *Solution*

continued

A $(E \cap F^c) \cap (E^c \cap F) = \varnothing$, and, moreover, by the formula for the probability of the complement of an event, $P(E^c) = P(F^c) = 1 - 0.6 = 0.4$. Hence

$$P(\text{one head}) = P((E \cap F^c) \cup (E^c \cap F))$$

$$= P(E) \cdot P(F^c) + P(E^c) \cdot P(F)$$

= (0.6)(0.4) + (0.4)(0.6)

= 0.48 = 48%.

c. What is the probability of obtaining no heads?

c. The event of obtaining no heads is $E^c \cap F^c$.

Example 9.9.6 – *Solution*

continued

Thus,

$$P(\text{no heads}) = P(E^c \cap F^c) = P(E^c) \cdot P(F^c)$$

$$= (0.4)(0.4) = 0.16 = 16\%.$$

d. What is the probability of obtaining at least one head?

d. There are two ways to solve this problem. One is to observe that because the event of obtaining one head and the event of obtaining two heads are mutually disjoint,

P(at least one head) = P(one head) + P(two heads)

Example 9.9.6 – *Solution*

continued

$= 0.48 + 0.36$ by parts (a) and (b)

$= 0.84$

$= 84\%$.

The second way is to use the fact that the event of obtaining at least one head is the complement of the event of obtaining no heads. So

$$
\begin{aligned}
P(\text{at least one head}) &= 1 - P(\text{no heads}) \\
&= 1 - 0.16 \quad \text{by part (c)} \\
&= 0.84 = 84\%.
\end{aligned}
$$

Example 9.9.7 – *Expected Value of Tossing a Loaded Coin Twice*

Suppose that a coin is loaded so that the probability of heads is 0.6, and suppose the coin is tossed twice.

If this experiment is repeated many times, what is the expected value of the number of heads?

Example 9.9.7 – *Solution*

Think of the outcomes of the coin tosses as just 0, 1, or 2 heads.

Example 9.9.6 showed that the probabilities of these outcomes are 0.16, 0.48, and 0.36, respectively.

Thus, by definition of expected value, the

$$\begin{aligned}\text{expected number of heads} &= 0 \cdot (0.16) + 1 \cdot (0.48) + 2 \cdot (0.36) \\ &= 1.2.\end{aligned}$$

Independent Events

Four conditions must be included in the definition of independence for three events.

Definition

Let A, B, and C be events in a sample space S. A, B, and C are **pairwise independent** if, and only if, they satisfy conditions 1–3 below. They are **mutually independent** if, and only if, they satisfy all four conditions below.

1. $P(A \cap B) = P(A) \cdot P(B)$
2. $P(A \cap C) = P(A) \cdot P(C)$
3. $P(B \cap C) = P(B) \cdot P(C)$
4. $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

Independent Events

The definition of mutual independence for any collection of n events with $n \geq 2$ generalizes the two definitions given previously.

Definition

Events $A_1, A_2, A_3, \ldots, A_n$ in a sample space S are **mutually independent** if, and only if, the probability of the intersection of any subset of the events is the product of the probabilities of the events in the subset.

Example 9.9.9 – *Tossing a Loaded Coin Ten Times*

A coin is loaded so that the probability of heads is 0.6 (and thus the probability of tails is 0.4). Suppose the coin is tossed ten times. As in Example 9.9.6, it is reasonable to assume that the results of the tosses are mutually independent.

a. What is the probability of obtaining eight heads?

b. What is the probability of obtaining at least eight heads?

Example 9.9.9 – *Solution*

a. For each $i = 1, 2, \ldots, 10$, let H_i be the event that a head is obtained on the i th toss, and let T_i be the event that a tail is obtained on the ith toss.

Suppose that the eight heads occur on the first eight tosses and that the remaining two tosses are tails. This is the event

$$H_1 \cap H_2 \cap H_3 \cap H_4 \cap H_5 \cap H_6 \cap H_7 \cap H_8 \cap T_9 \cap T_{10}.$$

For simplicity, we denote it as *HHHHHHHHTT*.

Example 9.9.9 – *Solution*

continued

By definition of mutually independent events,

$$P(HHHHHHHHTT) = (0.6)^8(0.4)^2.$$

Because of the commutative law for multiplication, if the eight heads occur on any other of the ten tosses, the same number is obtained. For instance, if we denote the event $H1 \cap H2 \cap T3 \cap H4 \cap H5 \cap H6 \cap H7 \cap H8 \cap T9 \cap H10$ by $H\ H\ T\ H\ H\ H\ H\ H\ T\ H$, then

$$P(HHTHHHHHTH) = (0.6)^2(0.4)(0.6)^5(0.4)(0.6) = (0.6)^8(0.4)^2.$$

Example 9.9.9 – *Solution*

continued

Now there are as many different ways to obtain eight heads in ten tosses as there are subsets of eight elements (the toss numbers on which heads are obtained) that can be chosen from a set of ten elements.

This number $\binom{10}{8}$. It follows that, because the different ways of obtaining eight heads are all mutually exclusive,

$$P(\text{eight heads}) = \binom{10}{8}(0.6)^8(0.4)^2.$$

Example 9.9.9 – *Solution*

continued

b. By reasoning similar to that in part (a),

$$P(\text{nine heads}) = \left[\begin{array}{l}\text{the number of different}\\ \text{ways nine heads can be}\\ \text{obtained in ten tosses}\end{array}\right] \cdot (0.6)^9(0.4)^1$$

$$= \binom{10}{9}(0.6)^9(0.4),$$

and

$$P(\text{ten heads}) = \left[\begin{array}{l}\text{the number of different}\\ \text{ways ten heads can be}\\ \text{obtained in ten tosses}\end{array}\right] \cdot (0.6)^{10}(0.4)^0$$

Example 9.9.9 – *Solution*

continued

$$= \binom{10}{10}(0.6)^{10}.$$

Because obtaining eight, obtaining nine, and obtaining ten heads are mutually disjoint events,

P(at least eight heads) = P(eight heads) + P(nine heads) + P(ten heads)

$$= \binom{10}{8}(0.6)^8(0.4)^2 + \binom{10}{9}(0.6)^9(0.4) + \binom{10}{10}(0.6)^{10}$$

Example 9.9.9 – *Solution*

continued

$$\cong 0.167$$

$$= 16.7\%.$$

Independent Events

Probabilities of the form

$$\binom{n}{k} p^{n-k}(1-p)^k,$$

where $0 \leq p \leq 1$, are called **binomial probabilities**.